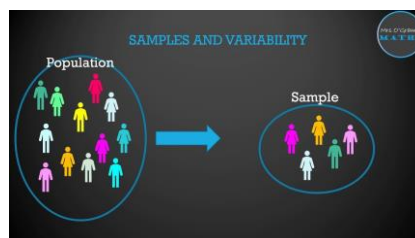


Inferential Statistics

Hisham A. Nematalla, PhD, PharmD

Inferential Statistics

- The objective of **inferential statistics** is to make **inferences** about **population parameters** based on **information** contained in a **sample**.
- **Sample measures** (i.e., **statistics**) are used to estimate **population measures** (i.e., **parameters**). These **statistics** are called **estimators**.



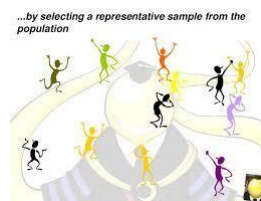
Inferential Statistics

- Inferential Statistics:
- Instead of using the entire population to gather the data, **a sample or samples** from the population will be collected to describe and make inferences about the entire population using only this sample.
- Properties of samples, such as the **mean or standard deviation**, are not called parameters, but **statistics**.

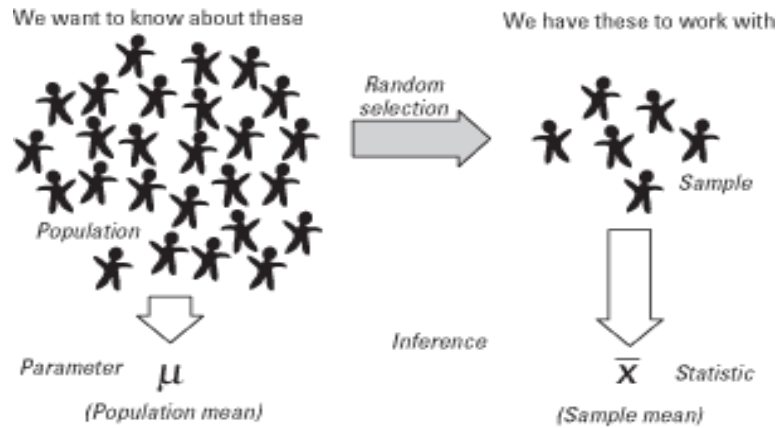
Inferential Statistics

Inferential statistics are valuable when

- it is **not convenient or possible** to examine each member of an entire population. (time , money)
- **no access** to the whole population you are interested in investigating, but only a limited number of data instead.
- make generalizations about the populations



Inferential Statistics

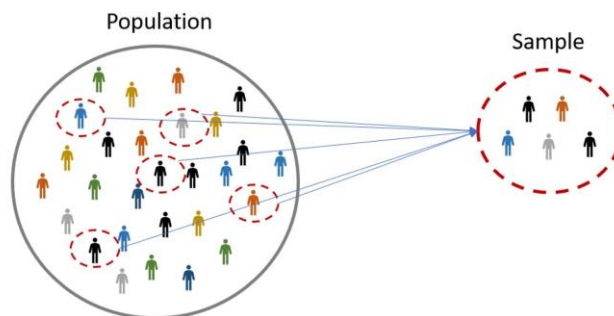


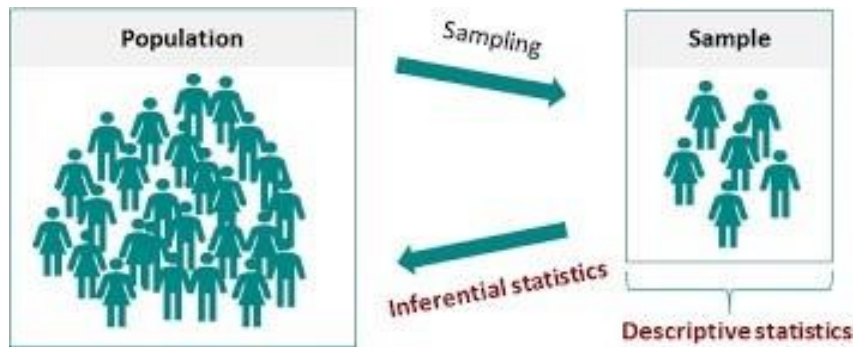
Inferential Statistics

	Population parameter	Sample Statistic
mean	μ	$\bar{x}$
Standard deviation	σ	s, sd

- **Sampling variability** is the tendency for a **statistic to vary** when computed on different samples from the same population. Each sample is a snapshot of a portion of the population.
- **Sampling variability** is different from the variability that we measure by computing a **sample variability (variance)**, which measures the amount of spread in a sample's scores.
- Sampling variability is the variation that we could expect in the many numeric values of a statistic that could be computed on repeated samples from the same population.

- What do we do with this knowledge that different samples would give us different results? **In other words**, how can researchers account for sampling variability in their statistics when typically they will study only one sample at a time?



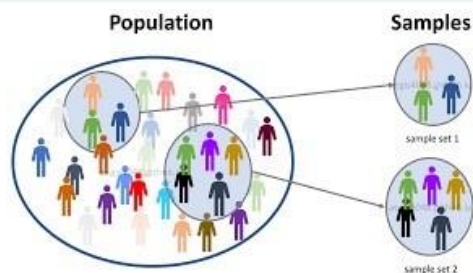


Check Your Understanding

7-1. What is the difference between *sample* variability and *sampling* variability?

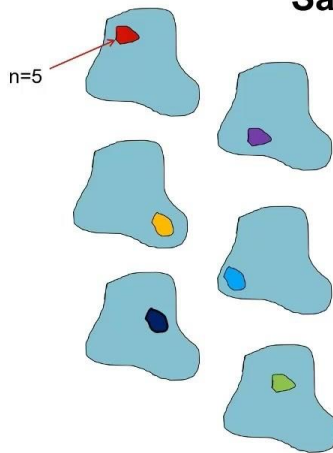
Suggested Answers

7-1. The term “*sample variability*” refers to the amount of spread in scores in a sample; the scores have variation in the sample. The term “*sampling variability*” refers to the amount of spread in the numeric values of a statistic computed on many samples from the same population; the numeric values of the statistic computed on multiple samples would vary from each other.

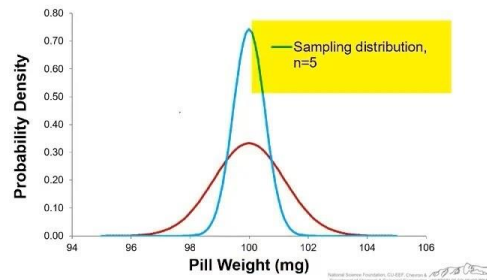


Sample distribution vs Population distribution

Sampling Distribution



1. Take sample of size n
2. Compute $\bar{X}$ of those n items
3. Repeat many times to generate a distribution of $\bar{X}$



A sampling distribution, which is a distribution of a statistic that could be formed by taking all possible samples of the same size from the same population, computing the same statistic on each sample, then arranging those numeric values of the statistic in a distribution along a number line.

Why We Need Sampling Distributions?

- We have implied why we need sampling distributions, but let's be more explicit about the reasons given so far, using the sample mean, M , as an example of a statistic with a sampling distribution:
- We often want **to generalize** from our known sample mean, M , to a hypothesized value of the population mean, μ .
- We know that **statistics have sampling variability**. The values for a sample mean will be different for the many different samples.
- **We want to know how typical our particular sample mean is compared with all possible sample means we could have computed.**
- If we wish **to compare our sample mean to a hypothesized population mean**, then we need to take into account the statistic's sampling variability.



Central Limit Theorem

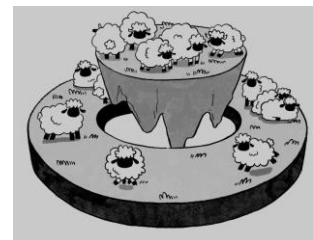
- With a large enough sample size and independent observations, the **sample mean will have a sampling distribution that follows a normal distribution**.
- Further, we could average together the pile of sample means from repeated **samples—the mean of the means**, if you will.
- The **Central Limit Theorem** says that **the mean of the means will equal the mean of the population from which we sampled**.
- We also could take the pile of sample means and compute **a measure of spread**.
- But we don't have to—the Central Limit Theorem says **the variance of the sampling distribution of M will equal the population variance divided by the sample size, or σ^2/N** .

Central Limit Theorem Formula



Sample mean = Population mean = μ

$$\begin{aligned}\text{Sample standard deviation} &= \frac{(\text{Standard deviation})}{\sqrt{n}} \\ &= \frac{\sigma}{\sqrt{n}}\end{aligned}$$



Methods for making inferences about parameters fall into one of two categories.

1.Estimation Procedures: **estimate** the value of the population parameter of interest.

2.Hypothesis Testing: **test a hypothesis** about the value of the parameter. These two methods of statistical inference: **estimation** and **hypothesis testing** involve different procedures, and, more important, they answer two different questions about the parameter.

In **estimating** a population parameter, we are answering the question, “What is the value of the population parameter?”

In **testing a hypothesis**, we are seeking an answer to the question, “Does the population parameter satisfy a specified condition?” For example, “ $\mu > 20$.”

Inferences About a Population Mean:

Large Sample Results

I-Estimation Procedures

Statistical test that will answer the question,

“What is the value of the population parameter?”

Estimation procedures can be divided into **two types**:

1. **Point Estimate:** We can calculate a **single number**, called a **point estimate**, from the sample data and use this value to estimate the parameter of interest.
2. **Interval Estimate:** Or we can compute **two numbers** from the sample data and use the interval formed by the two numbers as an **interval estimate** of the parameter of interest.

The **distinction** between the two is quite simple:

- In the first case we **compute one number and infer that the parameter is that number**.
- In the second case we compute **two numbers and infer that the parameter lies in the interval between them**.

- **Note:** The sample size **n** is assumed to be **30 or more**.

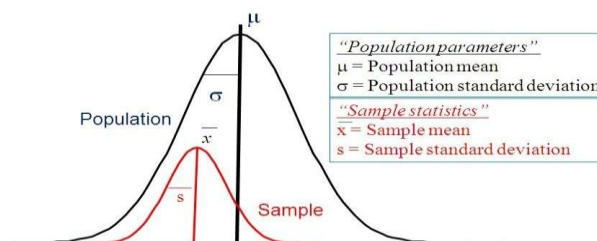
Estimation

- In statistics, **estimation** refers to the process by which one makes inferences about a population, based on information obtained from a sample.
- An estimate of a population parameter may be expressed in two ways:
 - Point Estimate
 - Interval Estimate

GUESSING

Point Estimate

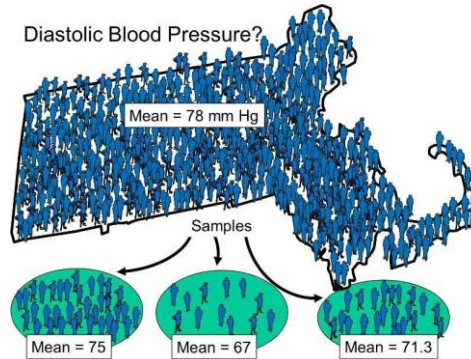
- A point **estimate** of a population parameter is a single value of a **statistic**.
 - For example, the **sample mean $\bar{x}$** is a point **estimate** of the **population mean μ** .



Point Estimate

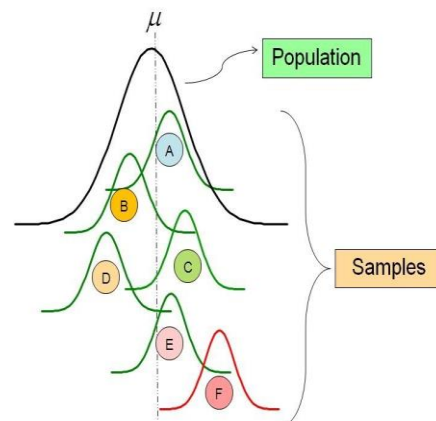
- Disadvantages:

- Point Estimates are not exact and generally vary from one sample to another.



Point Estimate

- the sample mean has some variability around the true population mean, which can be quantified using the standard deviation of this distribution of sample means.
- This tells us how far the typical estimate is away from the actual population mean.
- It also describes the typical error of the point estimate, and for this reason we usually call this standard deviation the **standard error (SE)** of the estimate.



Standard Error of the Mean

- Given n independent observations from a population with standard deviation ,
- the standard error of the sample mean is equal to: quantified
- the uncertainty of the sample mean using what we call the standard error,

$$SE = \frac{\sigma}{\sqrt{n}}$$

Where:

n = number of observations in a sample

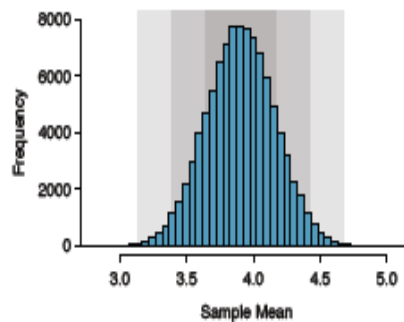
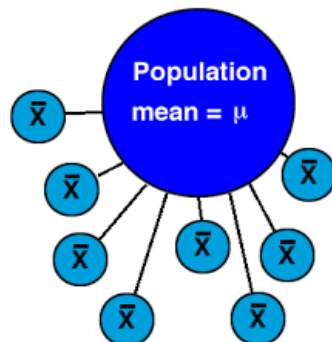
σ = standard deviation of the population

standard error $SE = \frac{\sigma}{\sqrt{n}}$ sample size

Example:
 $n = 5$
 $\sigma = 17$
 $= \frac{17}{\sqrt{5}}$
 $SE = 7.6$

Sampling Distribution

- It is useful to think of a particular point estimate as being drawn from such a distribution.



Central Limit Theorem

- we can conclude the distribution of sample means is nearly normal. This result can be explained by the Central Limit Theorem.
- Central Limit Theorem, informal description: If a sample consists of at least 30 independent observations and the data are not strongly skewed, then the distribution of the sample mean is well approximated by a normal model.



1. Point Estimation of a Population Mean, μ

Point estimate of μ : $\bar{Y}$

Bound on the error of estimation:

$$2\sigma_{\bar{Y}} = 2\sigma / \sqrt{N}$$

Note: The sample size n is assumed to be 30 or more

1. Point Estimation of a Population Mean, μ

A random sample of $n=150$ patients records indicate that the average Hb is **10.2 gm/dl**, with **standard deviation** of **1.5 mg/dl**. Use the sample information to **estimate μ** , the mean Hb for all patient on record on the hospital. **Place a bound on the error of estimation.**

$$\bar{y} = 10.2 \quad \text{and} \quad SD = 1.5$$

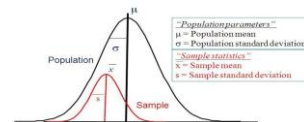
Hence, our **point estimate of μ** is **10.2.**, **Bound on the error of estimation:**

$$2\sigma_{\bar{y}} = 2\sigma / \sqrt{N}$$

When $n \geq 30$, we can replace σ by s

$$2\sigma_{\bar{y}} = 2s / \sqrt{n} = 2(1.5) / \sqrt{150} = 0.18$$

Knowing that approximately 95% of the sample means calculated in repeated sampling would lie within two standard deviations of μ , we feel fairly certain that $\bar{y} = 10.2$ is **within a distance of 0.18** of the **actual mean**.



Interval Estimate

- However, a point estimate is rarely perfect; usually there is some error in the estimate, i.e. by reporting a point estimate, we probably will not hit the exact population parameter.
- Instead of supplying just a point estimate of a parameter, a range of values for the parameter is provided (**=confidence interval**).



Confidence limits of the mean for large samples

- Using a **sample** to define a **range** which we may reasonably assume includes the **true value**. Such a range is known as a **confidence interval** and the extreme values of the interval are called the **confidence limits**.
- The term '**confidence**' implies that we can assert with a given **degree of confidence**, i.e. a **certain probability**, that the **confidence interval** does include the **true value**. The size of the confidence interval will obviously depend on how certain we want to be that it includes the true value: the greater the certainty, the greater the interval required.

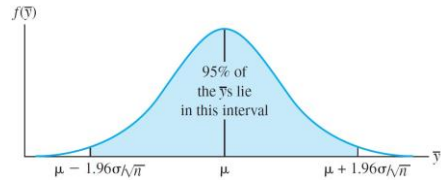
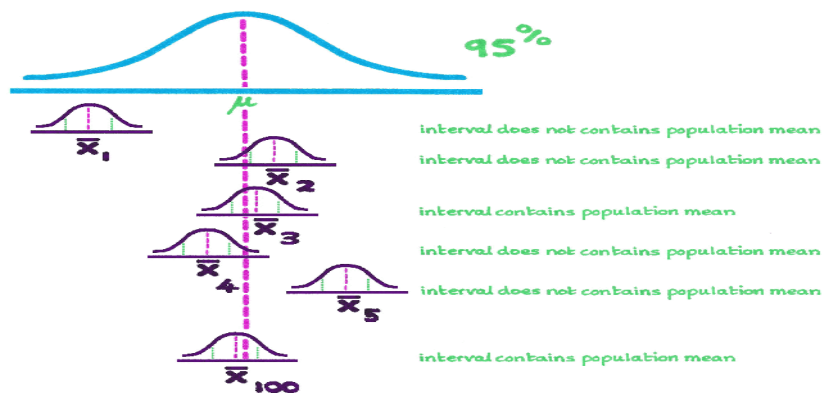


Figure 1 Sampling distribution for the mean, $\bar{y}$ showing the range within which 95% of sample means lie.

Figure 1 shows the sampling distribution of the mean for samples of size n . If we assume that this distribution is normal, then 95% of the sample means, $\bar{y}$ will lie in the range given by:

$$\mu - 1.96(\sigma/\sqrt{n}) < \bar{x} < \mu + 1.96(\sigma/\sqrt{n})$$

2
7



Approximately 95 intervals out of 100 created intervals will contain the unknown population mean

- We are XX% confident that the population parameter is between...
- Or about 95% of those intervals would contain the actual mean, μ



Interval Estimate

- The standard error represents the standard deviation associated with the estimate, and roughly 95% of the time the estimate will be within 2 standard errors of the parameter.
- If the interval spreads out 2 standard errors from the point estimate, we can be roughly 95% confident that we have captured the true parameter:



Interval Estimate

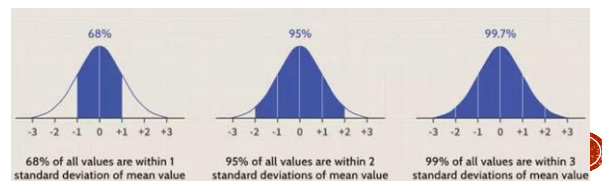
- There are three components to the confidence interval:

$$\text{point estimate} \pm 1.96 \times SE$$

- the point estimate,
- “1.96”,
- and the standard error.

- 1.96 corresponds to a 95% confidence level.

Confidence level	Z value
90%	1.65
95%	1.96
99%	2.58



Interval Estimate

Confidence interval for any confidence level

If the point estimate follows the normal model with standard error SE , then a confidence interval for the population parameter is

$$\text{point estimate} \pm z^* SE$$

where z^* corresponds to the confidence level selected.

Margin of error

In a confidence interval, $z^* \times SE$ is called the margin of error.



Conditions to use z score

Conditions for $\bar{x}$ being nearly normal and SE being accurate

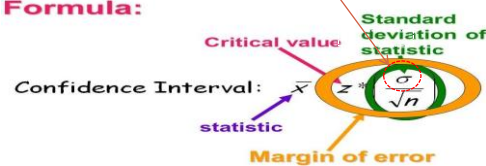
Important conditions to help ensure the sampling distribution of $\bar{x}$ is nearly normal and the estimate of SE sufficiently accurate:

- The sample observations are independent.
- The sample size is large: $n \geq 30$ is a good rule of thumb.
- The population distribution is not strongly skewed. This condition can be difficult to evaluate, so just use your best judgement.

Standard Error of the Mean

- the population standard deviation is typically unknown.
- we can use the point estimate of the standard deviation from the sample.

Formula:



Confidence limits of the mean for large samples, cont.

- ◆ (The exact value **1.96** has been used in this equation rather than the **approximate value, 2**, quoted under the **Empirical Rule**. You can use **Table 1 (Appendix)** to check that the proportion of values between $z = -1.96$ and $z = 1.96$ is indeed 0.95.)
- ◆ In practice, however, we usually have one sample, of known mean, and we require a range for μ , the true value.
- ◆ Equation $\mu - 1.96(\sigma/\sqrt{n}) < \bar{x} < \mu + 1.96(\sigma/\sqrt{n})$ can be rearranged to give this:

$$\bar{x} - 1.96(\sigma/\sqrt{n}) < \mu < \bar{x} + 1.96(\sigma/\sqrt{n})$$
- ◆ The above equation gives the **95% confidence interval of the true mean, μ** . The **95% confidence limits** are $\bar{x} \pm 1.96\sigma/\sqrt{n}$. In practice we are unlikely to know σ exactly. However, provided that the **sample is large**, σ can be replaced by its estimate, **s**.
- ◆ Other confidence limits are sometimes used, in particular the 99% and 99.7% confidence limits.

Consider the interval $\bar{Y} \pm 1.96\sigma_{\bar{Y}}$. Any time $\bar{Y}$ lies in the interval $\mu \pm 1.96\sigma_{\bar{Y}}$, the interval $\bar{Y} \pm 1.96\sigma_{\bar{Y}}$ will contain the parameter μ (Fig.2), and this will occur with probability 0.95. The interval $\bar{Y} \pm 1.96\sigma_{\bar{Y}}$ represents an interval estimate of μ .

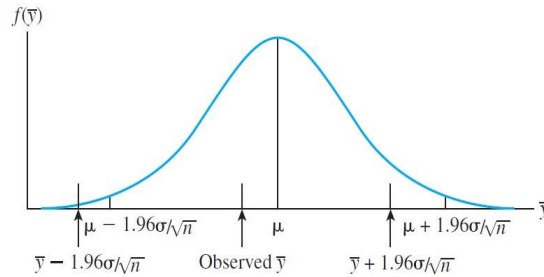


Figure 2 When the observed value of $\bar{Y}$ lies in the interval $\mu \pm 1.96\sigma_{\bar{Y}}$, the interval $\bar{Y} \pm 1.96\sigma_{\bar{Y}}$ contains the parameter μ .

10

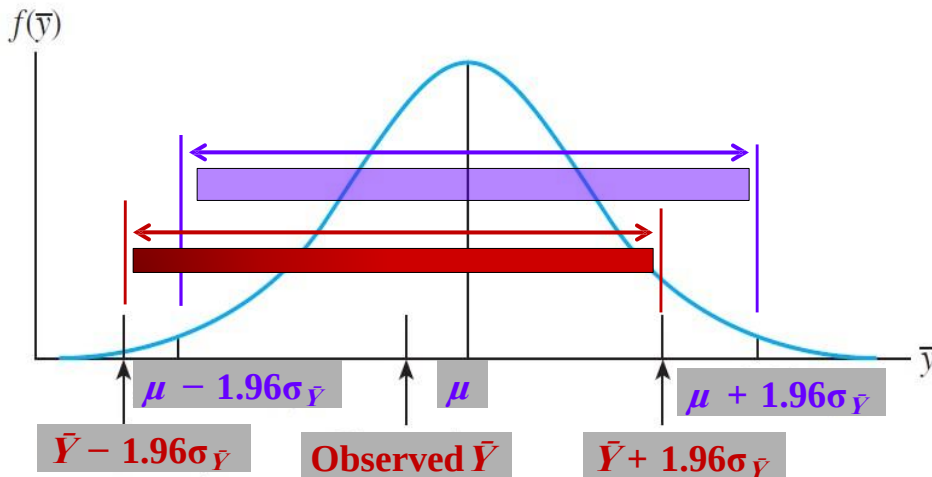


Figure 2 When the **observed value** of $\bar{Y}$ lies in the interval $\mu \pm 1.96\sigma_{\bar{Y}}$, the interval $\bar{Y} \pm 1.96\sigma_{\bar{Y}}$ contains the parameter μ .

11

Other confidence limits are sometimes used, in particular the 99% and 99.7% confidence limits. For large samples, the confidence limits of the mean are given by:

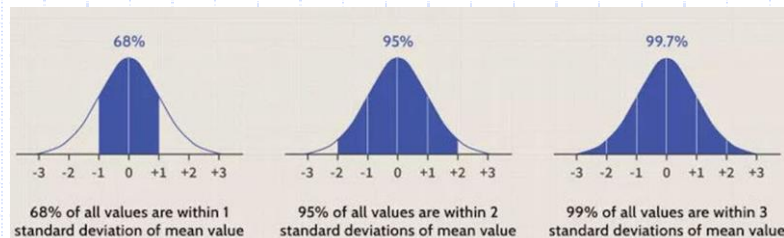
$$\bar{x} \pm zs/\sqrt{n}$$

where the value of z depends on the degree of confidence required.

For 95% confidence limits, $z = 1.96$

For 99% confidence limits, $z = 2.58$

For 99.7% confidence limits, $z = 2.97$



2. Interval Estimate

Confidence limits of the mean for large samples

- From sampling population, mean was found to be 6 and standard deviation = 0.5, $n=50$.
- Estimate the 95% and 99% confidence interval

$$\bar{x} - 1.96(\sigma/\sqrt{n}) < \mu < \bar{x} + 1.96(\sigma/\sqrt{n})$$

- In case of 99%, we use $z=2.58$ instead of 1.96
- In case of 99.7% we use $z= 2.97$.

II-Testing an Hypothesis About μ

A Statistical Test for μ

- The second type of inference making procedure is a **statistical test** of an hypothesis that **make an inference** about a population parameter.
 - Statistical test that will answer the question, “Is the population mean equal to a specified value μ_0 ?” **or** “Does the population parameter satisfy a specified condition?” For example, “ $\mu > 20$.”
 - A statistical test is **based on the concept of proof by contradiction** and is composed of five parts listed below.
-
- Considering that a new instructor at AIU university wish to evaluate the academic performance of 2025 graduate assuming that the affordability of resources (campus, e-learning platforms, foreign academics...) would help to improve such performance. His T.A suggest him to take the average grade as a statistic. Accordingly the T.A took a **sample form the student registry (n= 200)** and find **average =79** with **standard deviation = 10**. The new instructor astonished by the sample statistic would furtherly evaluate if the population mean would be equal or higher than **82**.

Five Parts of a Statistical Test

1. **Research hypothesis** (also called the **alternate hypothesis**), denoted by H_a . The statement about μ that the researcher is attempting to support or detect with the data from the study is the research hypothesis, H_a .
2. **Null hypothesis**, denoted by H_0 . The particular value specified for μ is called its **null value** and is denoted by μ_0 .
3. **Test statistic**, denoted by T.S.
4. **Rejection region**, denoted by R.R.
5. Check assumptions and draw **conclusions**.

A Statistical Test for μ

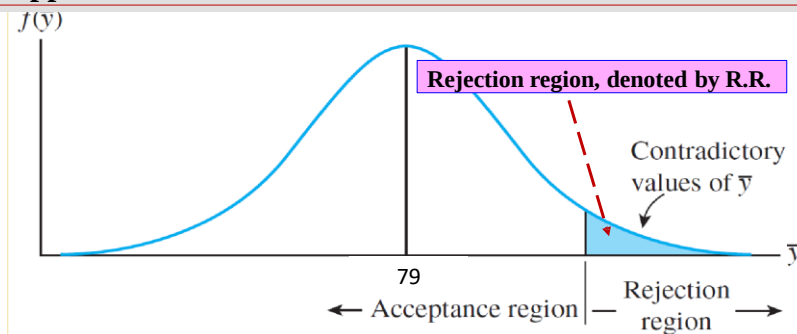
Five Parts of a Statistical Test concerning the **mean** grade for the student population

The **first step in setting up a statistical test** is determining the proper specification of H_0 and H_a .

1. **Research hypothesis, H_a** . We may be interested in the research hypothesis that the mean grade is above 82; that is, $H_a: \mu > 82$, the average observed in the past several years.
2. **Null hypothesis, denoted by H_0** . To verify the research hypothesis, we try to **contradict another hypothesis**, called the null hypothesis, $H_0: \mu \leq 79$ (i.e., the highest average yield that is still contradictory to the research hypothesis).

3. **Test statistic**, denoted by T.S. The decision to accept the null hypothesis or reject the null hypothesis in favor of the research hypothesis is based on T.S. computed from sample data (mean, $\bar{Y}$ and standard deviation, s).
3. **Rejection region**, denoted by R.R. Values of $\bar{Y}$ which are contradictory to the null hypothesis will lie in the upper tail of the distribution of $\bar{Y}$. See **Fig.1**. These contradictory values form a rejection region for our statistical test.
3. **Conclusion**. If the observed value of $\bar{Y}$ falls in the rejection region of **Fig.1**, **we would reject the null hypothesis that the mean $\mu \leq 79$** in favor of the research hypothesis that $\mu > 82$. Note that we are verifying the research hypothesis by contradicting the null hypothesis. If the observed value of $\bar{Y}$ falls in the acceptance region rather than the rejection region, we accept the null hypothesis.

FIGURE 1: Assuming that H_0 is true, contradictory values of $\bar{Y}$ are in the upper tail.



Conclusion. If the observed value of $\bar{Y}$ falls in the rejection region of **Fig.1**, we would reject the null hypothesis that the mean grade is $\mu = 79$ in favor of the research hypothesis that $\mu > 82$.

Note that we are verifying the research hypothesis by contradicting the null hypothesis. If the observed value of $\bar{Y}$ falls in the acceptance region rather than the rejection region, we accept the null hypothesis.

Alpha level (Alpha Error, α)

how the choice of α locates the rejection region.

- Alpha: The Level of Significance of a Test Alpha is

Alpha is also called Type I Error.

Alpha level: The probability of type I error is denoted by the symbol α .

A type I error (α) is committed if we reject the null hypothesis when it is true; i.e., α is the probability to wrongly reject the null hypothesis, H_0 .

- α is a preset value.

Specifying α : If we are willing to take the risk that 1 time in 20 we would incorrectly reject the null hypothesis, then $\alpha = 1/20 = 0.05$

Alpha level (Alpha Error, α)

- The value of **alpha** is associated with the **confidence level** of our test; e.g., for results with 95% level of confidence, the value of $\alpha = 1 - 0.95 = 0.05$ {since Confidence Level = $1 - \alpha$ }
- For a **95% confidence** level, **the value of alpha is 0.05**. The most common level for α is **5%** but it varies by application.
- This means that there is a **5% probability** that we will reject a **true null hypothesis**. In the long run, **one** out of every **twenty** hypothesis tests that we perform at this level will result in a **type I error**.

- ◆ The following lists some levels of confidence with their related values of **alpha**:
- ◆ For results with a 90% level of confidence, the value of **alpha** is $1 - 0.90 = \mathbf{0.10}$.
- ◆ For results with a 95% level of confidence, the value of **alpha** is $1 - 0.95 = \mathbf{0.05}$.
- ◆ For results with a 99% level of confidence, the value of **alpha** is $1 - 0.99 = \mathbf{0.01}$.
- ◆ Although in theory and practice many numbers can be used for **alpha**, the **most commonly used is 0.05**. **this level is appropriate in many cases, and historically, it has been accepted as the standard.**

Assuming the null hypothesis is true, then $\bar{Y}$ is normally distributed, with $\mu = 79$ and $\sigma_{\bar{Y}} = s / \sqrt{n} = 10 / \sqrt{200} = 0.7$. Since the **shaded area** of **Figure 2** corresponds to α , locating a rejection region with an area 0.025 in the right-hand tail of the distribution of $\bar{Y}$ would be equivalent to determining the value of z that has an area 0.025 to its right.

Referring to Table 1 Appendix, this value of $z=1.96$. Thus the rejection region for our example would be located 1.96 standard deviations ($1.96 \sigma_{\bar{Y}}$) above the mean $\mu = 79$. If the observed value of $\bar{Y}$ is greater than 1.96 standard deviations above the mean $\mu = 79$, we reject the null hypothesis as shown in **Fig.2**.

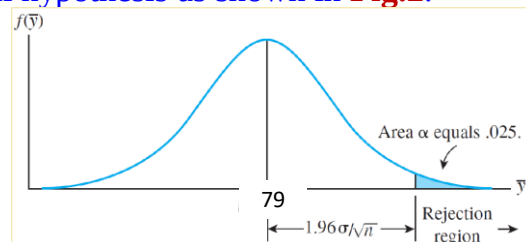
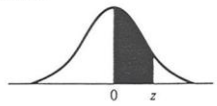


FIGURE 2: Rejection region for the grade example when $\alpha = .025$

Table 1 Normal curve areas



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990

Example 1: One-Tailed Test

Set up the parts of a statistical test for the grade example and use the sample data to reach a decision whether to accept or reject the null hypothesis. Set $\alpha = .025$.

$H_0: \mu \leq 79$

$H_a: \mu > 82$

T.S.: $\bar{Y}$

R.R.: For $\alpha = .025$, reject the null hypothesis if $\bar{Y}$ lies more than 1.96 standard errors above $\mu = 79$.

$$z = \frac{\bar{Y} - \mu_0}{\sigma / \sqrt{n}} = (82 - 79) / (10 / \sqrt{200}) = 2.121$$

Conclusion: since the observed value of $\bar{Y}$ lies more than 1.96 standard errors, in fact 2.121, standard errors above the hypothesized mean $\mu = 79$ we reject the null hypothesis in favor of the research hypothesis and conclude that the average grade is greater than 82.

The rejection region is located in only one tail of the distribution of $\bar{Y}$.

One-Tailed Test

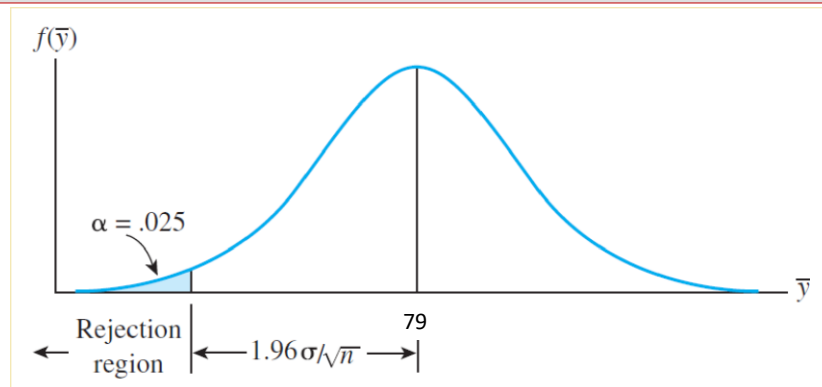
The statistical test conducted in previous Example is called a **one-tailed test because** the rejection region is located in only one tail of the distribution of $\bar{Y}$. If our research hypothesis is $H_a: \mu < 79$, small values of $\bar{Y}$ would indicate rejection of the null hypothesis. This test would also be one-tailed but the rejection region would be located in the lower tail of the distribution of $\bar{Y}$.

Figure 3 displays the rejection region for the alternative hypothesis $H_a: \mu < 79$, when $\alpha = .025$.

One-Tailed Test

FIGURE 3

Rejection region for $H_a: \mu < 79$ when $\alpha = .025$ for the grade example.



The rejection region is located in only one tail of the distribution of $\bar{Y}$